

MAT 310, Handout 20: Independent RVs and Expected Values (5.4-5.6)

Question: Given jointly distributed RVs Y_1 and Y_2 , how do you determine if they are independent?

Independence: Y_1 and Y_2 are independent if and only if, for all values of y_1 and y_2 ,

$$F(y_1, y_2) =$$

Equivalently, Y_1 and Y_2 are independent if and only if, for all values of y_1 and y_2 ,

- Discrete case: $p(y_1, y_2) =$
- Continuous case: $f(y_1, y_2) =$

Otherwise, Y_1 and Y_2 are dependent.

Corollary: If Y_1 and Y_2 are jointly distributed independent RVs, then for any intervals (a, b) and (c, d) ,

$$P(a < Y_1 < b, c < Y_2 < d) =$$

Example 1. Suppose Y_1 and Y_2 have the following joint density function:

$$f(y_1, y_2) = \begin{cases} 4y_1y_2, & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1 \\ 0, & o.w. \end{cases}.$$

Are Y_1 and Y_2 independent?

Example 2. Suppose Y_1 and Y_2 have the following joint density function:

$$f(y_1, y_2) = \begin{cases} 2, & 0 \leq y_2 \leq y_1 \leq 1 \\ 0, & o.w. \end{cases}.$$

Are Y_1 and Y_2 independent?

Theorem: Let Y_1 and Y_2 be jointly continuous RVs with joint density function $f(y_1, y_2)$ that is positive for $a \leq y_1 \leq b$ and $c \leq y_2 \leq d$, and 0 otherwise. Then, Y_1 and Y_2 are **independent** if and only if

$$f(y_1, y_2) = g(y_1)h(y_2)$$

for some nonnegative functions g and h .

Example 3. Suppose Y_1 and Y_2 have the following joint density function:

$$f(y_1, y_2) = \begin{cases} 2y_1, & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1 \\ 0, & o.w. \end{cases}.$$

Are Y_1 and Y_2 independent?

Expected Value of a Function of RVs:

- Given jointly discrete RVs $Y_1, Y_2, \dots, Y_k$ with joint probability function $p(y_1, y_2, \dots, y_k)$,

$$E[g(Y_1, Y_2, \dots, Y_k)] =$$

- Given jointly continuous RVs $Y_1, Y_2, \dots, Y_k$ with joint density function $f(y_1, y_2, \dots, y_k)$,

$$E[g(Y_1, Y_2, \dots, Y_k)] =$$

Example 4. Given jointly distributed continuous RVs Y_1 and Y_2 with joint density $f(y_1, y_2)$, set up integrals for the following:

(a) $E(Y_1) =$

(b) $E(Y_1 Y_2^2 + e^{Y_1}) =$

Example 5. Suppose Y_1 and Y_2 have the following joint density function:

$$f(y_1, y_2) = \begin{cases} 4y_1y_2, & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1 \\ 0, & o.w. \end{cases}.$$

Find $E(Y_1Y_2)$.

Properties of Expectation

1. $E(c) =$, where c is a constant.
2. $E[c \cdot g(Y_1, Y_2)] =$, where c is a constant and $g(Y_1, Y_2)$ is an arbitrary function.
3. $E \left[\sum_{i=1}^n g_i(Y_1, Y_2) \right] =$, where $g_i(Y_1, Y_2)$, $i = 1, \dots, n$, are arbitrary functions.
4. Let Y_1 and Y_2 be independent, $g(Y_1)$ be a function of Y_1 , and $h(Y_2)$ be a function of Y_2 . Then
$$E[g(Y_1)h(Y_2)] =$$

Proof (continuous case):

Example 6. Given independent $X \sim U(0, 1)$ and $Y \sim U(0, 2)$, find $E(XY^2)$.